

RS-232 Debug Box

Introduction

The RS232 Debug Box is a device that enables performing specific actions during an RS232 connection between two machines.

The box has three RS232 connectors: two female (a) and one male (b)



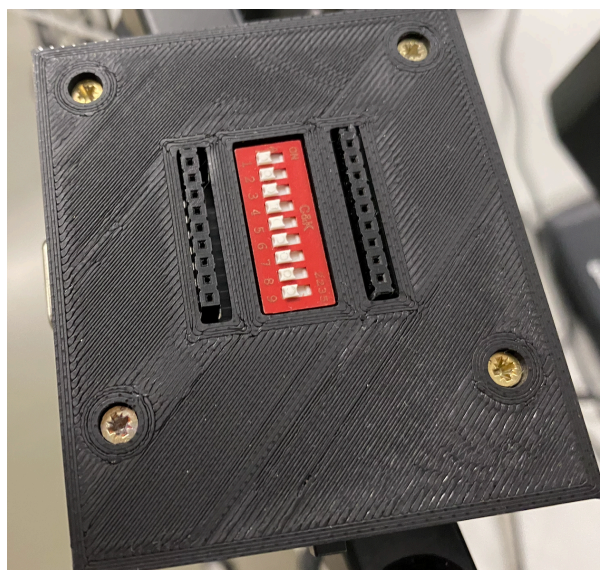
a)



b)

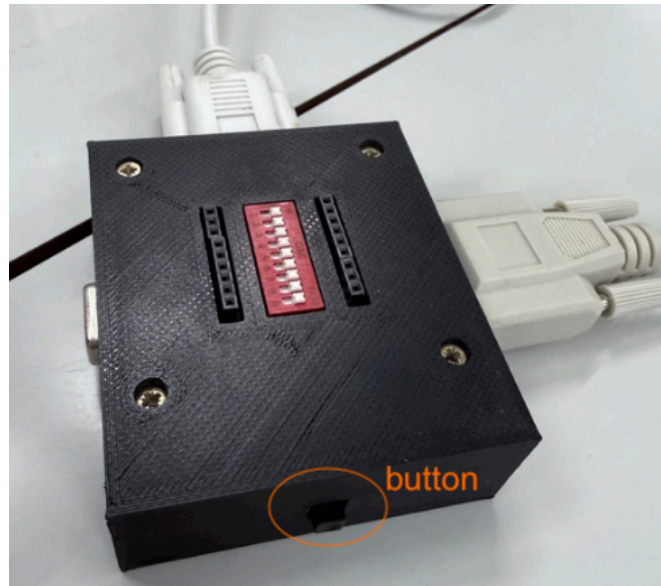
The cable on the male connector should not be removed.

The box offers dip switches, headers, and buttons, through which it is possible to interact with the serial transmission connection.



Normal connection and interruptions

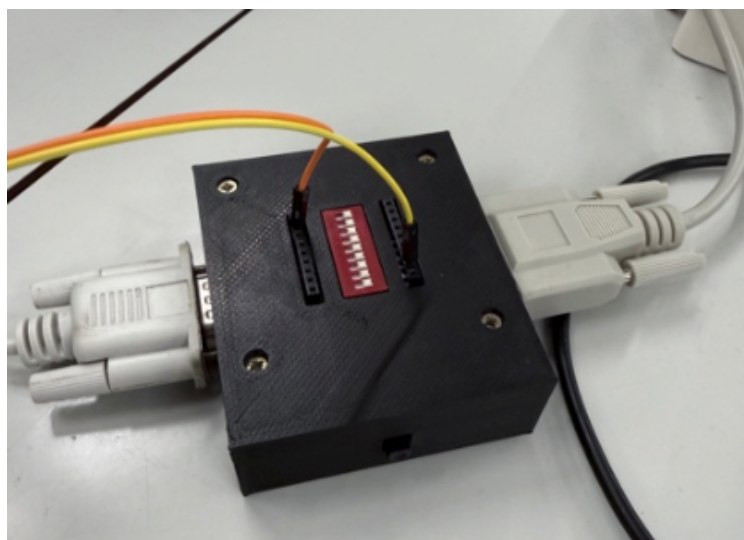
To test a normal connection, as well as to introduce interruptions in the connection, it is necessary to have the serial cables connected like this:



In this position, by pressing the button, the connection and data transmission will be interrupted, which will allow time-outs to occur; to reestablish the connection and resume transmission, you must release the button, restoring the initial conditions.

Introduction of errors in the communication

If you change the cable that is connected to the female plug to the other female plug available in the box, obtaining the configuration illustrated below, you may use the headers to introduce errors in the data that is being transmitted with the help of a cable, as represented below.



The ends of each color cable should be inserted into header 2 and header 7 on opposite sides, and whenever you want to introduce errors, simply make the other ends of the cables touch. We recommend testing this configuration for introducing errors with GTKterm, experimenting with sending data in both directions (by continuously pressing a specific keyboard letter alternately in each terminal and observing for errors).

Interruption of the connection in one single sense

It is possible to use the dip switches to interrupt the transmission of data in a single sense. To achieve this, it is necessary to have the cables connected as illustrated below and change the position of switch 2 or 3, depending on whether the objective is to interrupt the data transmission from the transmitter to the receiver or vice versa, as well as on which computer each code is running. Again, we recommend using GTKTerm to identify the proper switch for the intended purpose.

